

Java Data Structures

Complete Course — Zero to Advanced

JAVA 17+ | 16 DATA STRUCTURES | PHASE 1 → 4 | INTERVIEW READY

PHASE	LEVEL	TOPICS
1	Foundation	Memory, Big O, Generics, Java Basics
2	Beginner	Arrays, Strings, ArrayList, Stack, Queue, Deque
3	Intermediate	HashMap, LinkedList, Binary Tree, BST, Heap
4	Advanced	AVL Tree, Trie, Graph BFS/DFS, Segment Tree, Union-Find

How to use: Study each phase in order. Every topic covers: Concept → Visual → Complexity → Java Code → Dry Run → Edge Cases → Practice. Never skip phases — each one builds on the previous.

1.1 What is a Data Structure?

Definition: A data structure organises data in memory so that operations (access, search, insert, delete) are as fast as possible. Choosing the right one is often the difference between a solution that runs in 1 second vs 10 hours.

- **Array** — numbered shelf — fast $O(1)$ access by index
- **Stack** — pile of plates — LIFO: last added, first removed
- **Queue** — cinema line — FIFO: first added, first removed
- **HashMap** — dictionary — $O(1)$ key-to-value lookup
- **Tree** — org chart — hierarchical relationships
- **Graph** — road map — connections between any nodes

1.2 Memory: Stack vs Heap in Java

Java › Memory Model

[illegible]

1.3 Big O Notation

Big O describes how runtime grows as input size n increases. Rule: drop constants and keep only the dominant term. $O(2n) \rightarrow O(n)$, $O(n^2 + n) \rightarrow O(n^2)$.

Notation	Name	Example	At n = 1,000
$O(1)$	Constant	arr[0] / map.get(k)	1 step
$O(\log n)$	Logarithmic	Binary Search, BST ops	~10 steps

$O(n)$	Linear	Single for-loop	1,000 steps
$O(n \log n)$	Log-linear	Merge Sort, Heap Sort	~10,000 steps
$O(n^2)$	Quadratic	Nested loops	1,000,000 steps
$O(2^n)$	Exponential	Recursive subsets	Astronomical

Java › Big O Examples in Java

```
// O(1) – same time regardless of input size
int first = arr[0];
// O(n) – time grows linearly with n
for (int i = 0; i < n; i++) {
    process(arr[i]);
}
// O(n²) – avoid for large n (n=10,000 = 100 million ops)
for (int i = 0; i < n; i++) {
    for (int j = 0; j < n; j++) {
        compare(arr[i], arr[j]);
    }
}
// O(log n) – input halves each step
int lo = 0, hi = n - 1;
while (lo <= hi) {
    int mid = lo + (hi - lo) / 2; // avoids integer overflow
    if (arr[mid] == target) return mid;
    else if (arr[mid] < target) lo = mid + 1;
    else hi = mid - 1;
}
```

1.4 Java Generics & Comparable

Java › Generics — Write Once, Work for Any Type

```
// Without generics – you'd write separate IntBox, StringBox, etc.
```

```
class Box {  
    private T value;  
    public void set(T v) { this.value = v; }  
    public T get() { return value; }  
}
```

```
// Usage
```

```
Box intBox = new Box<>();
```

```
intBox.set(42);
```

```
int val = intBox.get(); // no cast needed
```

```
Box strBox = new Box<>();
```

```
strBox.set("hello");
```

```
// Comparable – allows Collections.sort() to work on your class
```

```
class Student implements Comparable {
```

```
    String name;
```

```
    int gpa;
```

```
@Override
```

```
    public int compareTo(Student other) {
```

```
        return Integer.compare(this.gpa, other.gpa); // ascending
```

```
    }
```

```
}
```

```
List students = new ArrayList<>();
```

```
Collections.sort(students); // works because of Comparable
```

2.1 Arrays

Analogy: A row of numbered mailboxes. Jump to any box instantly ($O(1)$) using its index number. But inserting a new box in the middle requires shifting all boxes after it — $O(n)$.

Java › Arrays — 1D, 2D, and Dynamic Resizing

[illegible]

Operation	Array	ArrayList
Access by index	O(1)	O(1)
Search	O(n)	O(n)
Insert at end	O(1)	O(1) amortised
Insert at middle	O(n)	O(n)
Delete	O(n)	O(n)

- **Edge case:** `arr[-1]` or `arr[arr.length]` → `ArrayIndexOutOfBoundsException`
- **Edge case:** if `arr` is null → `arr.length` throws `NullPointerException`
- **Use `Array`** when size is known and fixed (slightly faster, less memory)
- **Use `ArrayList`** when size changes at runtime

2.2 Stack

Analogy: A stack of plates. You can only add (push) or remove (pop) from the TOP. Last plate placed = first plate taken — **LIFO: Last In, First Out**.

Java › Stack — Custom Implementation Using Array

```
class Stack {
    private Object[] data;
    private int top = -1; // -1 means empty
    private int capacity;
    public Stack(int capacity) {
this.capacity = capacity;
this.data = new Object[capacity];
    }

    // Add element to top — O(1)
    public void push(T item) {
        if (top == capacity - 1) {
            throw new RuntimeException("Stack overflow");
        }
        data[++top] = item; // increment top FIRST, then assign
    }

    // Remove and return top element — O(1)
    @SuppressWarnings("unchecked")
    public T pop() {
        if (isEmpty()) {
            throw new RuntimeException("Stack underflow");
        }

        return (T) data[top--]; // return top, then decrement
    }

    // Look at top without removing — O(1)
    @SuppressWarnings("unchecked")
    public T peek() {
        if (isEmpty()) throw new RuntimeException("Stack is empty");
        return (T) data[top];
    }

    public boolean isEmpty() { return top == -1; }
    public int size() { return top + 1; }
}
```

Java › Stack — Java Built-in (Recommended)

```
// Always use Deque instead of the legacy Stack class
Deque stack = new ArrayDeque<>();
stack.push(10); // top → 10
stack.push(20); // top → 20
stack.push(30); // top → 30
stack.peek(); // 30 - O(1), no removal
stack.pop(); // 30 - O(1), removes top
stack.pop(); // 20
stack.isEmpty(); // false (10 still there)
```

Dry Run — push(10), push(20), push(30), pop()

Java › Dry Run — Stack Operations

```
// Initial state: data=[ _, _, _ ] top = -1
push(10) → data=[10, _, _] top = 0
push(20) → data=[10, 20, _] top = 1
push(30) → data=[10, 20, 30] top = 2
pop() → returns 30 top = 1 data=[10, 20, _]
peek() → returns 20 top = 1 (no change)
```

Real uses: Undo/Redo, browser back button, expression evaluation, DFS, call stack.

2.3 Queue & Dequeue

Queue: A cinema ticket line — **FIFO: First In, First Out**. **Deque** (double-ended queue) allows adding or removing from BOTH ends. It acts as both a Stack and a Queue.

Java › Queue & Deque

[illegible]

Real uses: BFS traversal, task scheduling, sliding window problems, print queue.

3.1 HashMap & HashSet

Analogy: A dictionary. You look up a word (key) and instantly get its definition (value). Internally a **hash function** maps each key to an array bucket index. Collisions (two keys $\rightarrow$ same index) are resolved by chaining (LinkedList at each bucket).

Java › HashMap — Key-Value Lookup in O(1)

```
// ■■ Basic Operations ████████████████████████████████████████████████████
```

```
HashMap scores = new HashMap<>();  
scores.put("Alice", 95);  
scores.put("Bob", 82);  
scores.put("Carol", 91);  
scores.get("Bob"); // 82 – O(1)  
scores.getDefault("Dave", 0); // 0 – safe fallback  
scores.containsKey("Alice"); // true – O(1)  
scores.remove("Carol"); // removes entry  
scores.size(); // 2  
  
// ■■ Iterate all entries ████████████████████████████████████████████████████  
for (Map.Entry entry : scores.entrySet()) {  
    System.out.println(entry.getKey() + " → " + entry.getValue());  
}  
  
// ■■ Count character frequencies (classic pattern) ██████████████████████  
  
String text = "hello world";  
Map freq = new HashMap<>();  
for (char c : text.toCharArray()) {  
    freq.put(c, freq.getDefault(c, 0) + 1);  
}  
  
// freq: {h=1, e=1, l=3, o=2, ' '=1, w=1, r=1, d=1}
```

Java › HashSet — Unique Elements, O(1) Membership

[illegible]

Operation	Average	Worst (all collide)
put	O(1)	O(n)
get	O(1)	O(n)
remove	O(1)	O(n)
containsKey	O(1)	O(n)

- **HashMap** — allows one null key, unordered, $O(1)$ average
- **TreeMap** — sorted by key, $O(\log n)$ for all ops, no null keys
- **LinkedHashMap** — insertion-order preserved, great for LRU Cache

3.2 LinkedList — Built from Scratch

Analogy: A treasure hunt. Each clue (Node) contains data and tells you where the next clue is. Unlike arrays, there is no random access — you must walk node by node from the head.

Java › Node Class (Building Block)

```
class Node {
    T data;
    Node next;
    Node(T data) {
        this.data = data;
        this.next = null;
    }
}
```

Java › Singly LinkedList — addFirst, addLast, delete

```

class SinglyLinkedList {
Node head;

    int size;
    // Add to front – O(1)
    public void addFirst(T data) {
Node newNode = new Node<>(data);
newNode.next = head;
head = newNode;
size++;
}

    // Add to end – O(n)
    public void addLast(T data) {
Node newNode = new Node<>(data);
        if (head == null) { head = newNode; size++; return; }
Node current = head;
        while (current.next != null) current = current.next;
current.next = newNode;
size++;
}

    // Delete by value – O(n)
    public boolean delete(T data) {
        if (head == null) return false;
        if (head.data.equals(data)) { head = head.next; size--; return true; }
Node current = head;
        while (current.next != null) {
            if (current.next.data.equals(data)) {
current.next = current.next.next;
size--; return true;
}
current = current.next;
}

        return false;
}
}

```

Java › Singly LinkedList — reverse() in-place

```

// Reverse in-place – O(n) time, O(1) space
public void reverse() {
Node prev = null;
Node current = head;
Node next = null;
    while (current != null) {
next = current.next; // 1. save next node
current.next = prev; // 2. reverse the pointer
prev = current; // 3. advance prev
current = next; // 4. advance current
}
head = prev; // prev is now the new head
}

```

Dry Run — Reverse: head → [1] → [2] → [3] → null

Java › Dry Run — Reverse LinkedList

```
// Before: head → [1] → [2] → [3] → null
// prev=null, current=[1], next=?
// Step 1: next=[2] [1].next=null prev=[1] current=[2]
// Step 2: next=[3] [2].next=[1] prev=[2] current=[3]
// Step 3: next=null [3].next=[2] prev=[3] current=null
// After: head = prev = [3]
// Result: head → [3] → [2] → [1] → null
```

3.3 Binary Tree & Traversals

Analogy: A company org chart. CEO is the root. Every node has at most 2 children (left and right). Leaf nodes have no children. Every node has exactly one parent (except root).

Java › TreeNode Class

```
// Tree structure:
// 1 <- root
// / \
// 2 3
// / \
// 4 5 <- leaves
class TreeNode {
    int val;
    TreeNode left;
    TreeNode right;
    TreeNode(int val) { this.val = val; }
}
```

Java › Binary Tree — Inorder, Preorder, Postorder

```
// INORDER: Left -> Root -> Right => 4 2 5 1 3
void inorder(TreeNode node) {
    if (node == null) return;
    inorder(node.left);
    System.out.print(node.val + " ");
    inorder(node.right);
}

// PREORDER: Root -> Left -> Right => 1 2 4 5 3
void preorder(TreeNode node) {
    if (node == null) return;
    System.out.print(node.val + " ");
    preorder(node.left);
    preorder(node.right);
}

// POSTORDER: Left -> Right -> Root => 4 5 2 3 1
void postorder(TreeNode node) {
    if (node == null) return;
    postorder(node.left);
    postorder(node.right);
    System.out.print(node.val + " ");
}
```

Java › Binary Tree — Level Order (BFS) + Height

```
// LEVEL ORDER – row by row using Queue => 1 2 3 4 5
void levelOrder(TreeNode root) {
    if (root == null) return;
    Queue q = new LinkedList<>();
    q.offer(root);
    while (!q.isEmpty()) {
        TreeNode node = q.poll();
        System.out.print(node.val + " ");
        if (node.left != null) q.offer(node.left);
        if (node.right != null) q.offer(node.right);
    }
}

// HEIGHT – O(n), uses recursion
int height(TreeNode node) {
    if (node == null) return 0;
    return 1 + Math.max(height(node.left), height(node.right));
}
```

3.4 Binary Search Tree (BST)

BST Rule: For every node: all values in the **left subtree** are smaller, all in the **right subtree** are larger. This halves the search space at each step — $O(\log n)$ for balanced trees.

Java › BST — Insert and Search

```

// BST rule: left < node < right
// 8
// / \
// 3 10
// / \ \
// 1 6 14

class BST {
    TreeNode root;

    // Insert - O(log n) avg, O(n) worst
    TreeNode insert(TreeNode node, int val) {
        if (node == null) return new TreeNode(val);
        if (val < node.val) node.left = insert(node.left, val);
        else if (val > node.val) node.right = insert(node.right, val);
        return node;
    }

    // Search - O(log n) avg
    boolean search(TreeNode node, int val) {
        if (node == null) return false;
        if (val == node.val) return true;
        return (val < node.val)
            ? search(node.left, val)
            : search(node.right, val);
    }

    TreeNode findMin(TreeNode node) {
        while (node.left != null) node = node.left;
        return node;
    }
}

```

Java › BST — Delete (3 cases)

```

TreeNode delete(TreeNode node, int val) {
    if (node == null) return null;
    if (val < node.val) node.left = delete(node.left, val);
    else if (val > node.val) node.right = delete(node.right, val);
    else {
        // Case 1: leaf - just remove
        if (node.left == null && node.right == null) return null;
        // Case 2: one child - return the surviving child
        if (node.left == null) return node.right;
        if (node.right == null) return node.left;
        // Case 3: two children
        // Replace value with inorder successor, then delete it
        TreeNode successor = findMin(node.right);
        node.val = successor.val;
        node.right = delete(node.right, successor.val);
    }

    return node;
}

```

Operation	Time	Notes
offer	$O(\log n)$	Insert + sift up
poll	$O(\log n)$	Remove root + sift down
peek	$O(1)$	Just read root
heapify (all)	$O(n)$	Build heap from existing array

4.1 AVL Tree — Self-Balancing BST

Problem with plain BST: Inserting sorted data [1,2,3,4,5] degrades a BST into a linked list — all operations become $O(n)$. An **AVL Tree** self-balances after every insert/delete using rotations, guaranteeing $O(\log n)$ for all operations in all cases.

Java › AVL Node Class

```
// Balance Factor (BF) = height(left) - height(right)
// AVL rule: BF must be -1, 0, or +1 at EVERY node
// If |BF| > 1 => rotate to restore balance
class AVLNode {
    int val;
    int height;
    AVLNode left;
    AVLNode right;
    AVLNode(int val) { this.val = val; this.height = 1; }
}

class AVLTree {
    private int height(AVLNode n) { return (n == null) ? 0 : n.height; }
    private int balanceFactor(AVLNode n) {
        return (n == null) ? 0 : height(n.left) - height(n.right);
    }
    private void updateHeight(AVLNode n) {
        n.height = 1 + Math.max(height(n.left), height(n.right));
    }
}
```

Java › AVL Tree — Right and Left Rotations

```

// Right Rotation – fixes Left-Left imbalance
// Before: z After: y
// / \ / \
// y T4 x z
// / \ / \
// x T3 T3 T4

private AVLNode rotateRight(AVLNode z) {
    AVLNode y = z.left;
    AVLNode T3 = y.right;
    y.right = z; // rotation
    z.left = T3;
    updateHeight(z);
    updateHeight(y);
    return y; // y is new subtree root
}

// Left Rotation – fixes Right-Right imbalance
private AVLNode rotateLeft(AVLNode z) {
    AVLNode y = z.right;
    AVLNode T2 = y.left;
    y.left = z;
    z.right = T2;
    updateHeight(z);
    updateHeight(y);
    return y;
}

```

```

AVLNode insert(AVLNode node, int val) {
    // Step 1: Standard BST insert
    if (node == null) return new AVLNode(val);
    if (val < node.val) node.left = insert(node.left, val);
    else if (val > node.val) node.right = insert(node.right, val);
    else return node; // no duplicates
    // Step 2: Update height
    updateHeight(node);
    // Step 3: Check balance and fix
    int bf = balanceFactor(node);
    // LL – single right rotation
    if (bf > 1 && val < node.left.val) return rotateRight(node);
    // RR – single left rotation
    if (bf < -1 && val > node.right.val) return rotateLeft(node);
    // LR – left-right double rotation
    if (bf > 1 && val > node.left.val) {
        node.left = rotateLeft(node.left);
        return rotateRight(node);
    }
    // RL – right-left double rotation
    if (bf < -1 && val < node.right.val) {
        node.right = rotateRight(node.right);
        return rotateLeft(node);
    }
    return node;
}

```

Operation	Plain BST (worst)	AVL Tree (always)
Search	$O(n)$	$O(\log n)$
Insert	$O(n)$	$O(\log n)$
Delete	$O(n)$	$O(\log n)$

4.2 Trie (Prefix Tree)

Analogy: Phone contact autocomplete. Each node represents one character. Words sharing a prefix share the same root-to-node path. Search time depends on word length (L), not total number of words — $O(L)$.

Java › TrieNode + Insert

```
// Visual: app, apple, apply
// root->[a]->[p]->[p*]->[l]->[e*]
// ->[y*] (* = word end)
class TrieNode {
TrieNode[] children = new TrieNode[26];
    boolean isEnd = false;
}
class Trie {
TrieNode root = new TrieNode();
    // Insert word - O(L)
    public void insert(String word) {
TrieNode current = root;
        for (char ch : word.toCharArray()) {
            int index = ch - 'a';
            if (current.children[index] == null)
current.children[index] = new TrieNode();
            current = current.children[index];
        }
        current.isEnd = true;
    }
}
```

```

// Search exact word — O(L)
public boolean search(String word) {
    TrieNode current = root;
    for (char ch : word.toCharArray()) {
        int index = ch - 'a';
        if (current.children[index] == null) return false;
        current = current.children[index];
    }
    return current.isEnd; // must be a complete word
}

// Check prefix exists — O(L)
public boolean startsWith(String prefix) {
    TrieNode current = root;
    for (char ch : prefix.toCharArray()) {
        int index = ch - 'a';
        if (current.children[index] == null) return false;
        current = current.children[index];
    }
    return true; // path exists — may or may not be a full word
}

// Usage
Trie trie = new Trie();
trie.insert("apple"); trie.insert("app"); trie.insert("apply");
trie.search("app"); // true
trie.search("ap"); // false — not inserted
trie.startsWith("app"); // true
trie.startsWith("xyz"); // false

```

4.3 Graph — BFS & DFS

Analogy: Google Maps. Cities are **vertices**. Roads are **edges**. **BFS** (Queue) finds the shortest path measured in hops. **DFS** (Stack/recursion) explores as deep as possible before backtracking.

Java › Graph — Adjacency List + BFS

```

// Build adjacency list (space-efficient)
int V = 6;
List< graph = new ArrayList<>();
for (int i = 0; i < V; i++) graph.add(new ArrayList<>());
void addEdge(int u, int v) {
graph.get(u).add(v);
graph.get(v).add(u); // undirected
}
// BFS — level by level using Queue — O(V + E)
void bfs(int start) {
boolean[] visited = new boolean[V];
Queue queue = new LinkedList<>();
queue.offer(start);
visited[start] = true;
while (!queue.isEmpty()) {
int node = queue.poll();
System.out.print(node + " ");
for (int nb : graph.get(node)) {
if (!visited[nb]) {
visited[nb] = true;
queue.offer(nb);
}
}
}
}
}

```

Java › Graph — DFS + Cycle Detection

```

// DFS — go deep first using recursion — O(V + E)
void dfs(int node, boolean[] visited) {
visited[node] = true;
System.out.print(node + " ");
for (int nb : graph.get(node)) {
if (!visited[nb]) dfs(nb, visited);
}
}
// Cycle Detection in undirected graph via DFS
// A back edge (to non-parent visited node) = cycle
boolean hasCycle(int node, int parent, boolean[] visited) {
visited[node] = true;
for (int nb : graph.get(node)) {
if (!visited[nb]) {
if (hasCycle(nb, node, visited)) return true;
} else if (nb != parent) {
return true; // back edge found
}
}
return false;
}
}

```

Algorithm	Time	Space	Best For
BFS	$O(V + E)$	$O(V)$	Shortest path (unweighted)
DFS	$O(V + E)$	$O(V)$	Cycle detect, topo sort
Dijkstra	$O((V+E) \log V)$	$O(V)$	Shortest path (weighted, non-negative)
Adjacency List	—	$O(V+E)$	Sparse graphs (preferred)
Adjacency Matrix	—	$O(V^2)$	Dense graphs, $O(1)$ edge check

4.4 Union-Find / Disjoint Set

Analogy: Friend circles in a social network. "Are Alice and Bob in the same circle?" Union-Find answers connectivity queries in nearly $O(1)$ using two key optimisations: **path compression** (flatten tree during find) and **union by rank** (attach smaller tree under taller one).

Java › Union-Find with Path Compression + Union by Rank

[illegible]

Operation	Time Complexity
find	$O(\alpha(n)) \approx O(1)$ — inverse Ackermann, practically constant
union	$O(\alpha(n)) \approx O(1)$

connected	$O(\alpha(n)) \approx O(1)$
------------------	-----------------------------

Practice Problems

[EASY] **Array** — Find the maximum element in an unsorted array.

Java › Solution — Array

```
int max = arr[0];
for (int x : arr) {
    max = Math.max(max, x);
}
return max; // O(n) time, O(1) space
```

[EASY] **Stack** — Check if a string's brackets are balanced: '({[]})' → true, '({[])' → false.

Java › Solution — Stack

```
Deque stack = new ArrayDeque<>();
for (char c : s.toCharArray()) {
    if (c == '(' || c == '{' || c == '[') {
        stack.push(c);
    } else {
        if (stack.isEmpty()) return false;
        char top = stack.pop();
        if (c == ')' && top != '(') return false;
        if (c == '}' && top != '{') return false;
        if (c == ']' && top != '[') return false;
    }
}
return stack.isEmpty(); // true only if all brackets matched
```

[EASY] **HashMap** — Count the frequency of each character in a string.

Java › Solution — HashMap

```
Map freq = new HashMap<>();
for (char c : s.toCharArray()) {
    freq.put(c, freq.getOrDefault(c, 0) + 1);
}
return freq; // O(n) time and space
```

[MEDIUM]
]

LinkedList — Reverse a singly linked list in-place in O(n) time, O(1) space.

Try solving this yourself first. Think: which data structure gives the right time complexity? Write the brute force first, then optimise.

[MEDIUM]
]

Binary Tree — Find the maximum depth of a binary tree.

Try solving this yourself first. Think: which data structure gives the right time complexity? Write the brute force first, then optimise.

[MEDIUM
]

HashMap — Two Sum: given array and target, return indices of two numbers that add up to target.

Try solving this yourself first. Think: which data structure gives the right time complexity? Write the brute force first, then optimise.

[HARD]

Graph — Detect a cycle in an undirected graph using DFS.

Try solving this yourself first. Think: which data structure gives the right time complexity? Write the brute force first, then optimise.

[HARD]

Heap — Find the K-th largest element in an unsorted array. Target: $O(n \log k)$.

Try solving this yourself first. Think: which data structure gives the right time complexity? Write the brute force first, then optimise.

[HARD]

Trie — Implement autocomplete: given a prefix, return all words in the Trie starting with it.

Try solving this yourself first. Think: which data structure gives the right time complexity? Write the brute force first, then optimise.

Quick Reference Cheat Sheet

When to Use Which Data Structure

Data Structure	Best For	Java Class
Array	Fixed-size, index access	int[], String[]
ArrayList	Dynamic array, random access	ArrayList
LinkedList	Frequent head insert / delete	LinkedList
Stack (LIFO)	Undo, DFS, expression eval	Deque (ArrayDeque)
Queue (FIFO)	BFS, scheduling, print queue	Queue (LinkedList)
PriorityQueue	Min/Max access, greedy problems	PriorityQueue
HashMap	O(1) key-value lookup	HashMap
TreeMap	Sorted keys, range queries	TreeMap
HashSet	Unique elements, fast membership	HashSet
BST	Sorted data, floor/ceiling queries	TreeSet
Trie	Prefix search, autocomplete	Custom
Graph	Connections, shortest paths	Custom (adj list)
Union-Find	Connectivity, Kruskal's MST	Custom

Big O Complexity at a Glance

Notation	Name	Example Operation	Speed
$O(1)$	Constant	arr[i], map.get(k), heap.peek()	Instant
$O(\log n)$	Logarithmic	Binary search, BST/AVL ops	Very fast
$O(n)$	Linear	Single loop, LinkedList search	Fast
$O(n \log n)$	Log-linear	Merge Sort, Heap Sort	Moderate
$O(n^2)$	Quadratic	Bubble Sort, nested loops	Slow for large n
$O(2^n)$	Exponential	Recursive subsets, naive DP	Avoid

Common Mistakes to Avoid

- **NullPointerException** — check head != null before traversal; check arr != null before arr.length
- **ArrayIndexOutOfBoundsException** — index must satisfy: $0 \leq i < \text{arr.length}$
- **ConcurrentModificationException** — never modify a collection while iterating with for-each

-
- **Integer overflow in mid** — use $\text{mid} = \text{lo} + (\text{hi} - \text{lo}) / 2$, NOT $(\text{lo} + \text{hi}) / 2$
 - **Stack overflow in DFS** — for large graphs, use iterative DFS with an explicit Stack
 - **Off-by-one in LinkedList** — stop when `current.next == null` (not `current == null`) to avoid NPE
 - **Wrong HashMap key type** — use `Integer`, not `int` (autoboxed); know that `equals()` checks value, not reference
 - **Modifying during stream** — collect results first, then apply changes to the original collection

Your next steps after completing this course: (1) Solve 100 LeetCode problems (40 Easy, 50 Medium, 10 Hard). (2) Learn the 5 core Sorting algorithms: Bubble, Selection, Insertion, Merge, Quick. (3) Study Dynamic Programming — it combines data structures with recursion. (4) Aim to code every structure from scratch in under 20 minutes without notes.